Perfume Price Prediction Using Machine Learning

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Project Report

1.Introduction

With the rapid growth of e-commerce platforms, online perfume shopping has become highly competitive. Customers compare products based on brand reputation, bottle size, ratings, discounts, and final selling price. For businesses, predicting the correct selling price is important for maximizing profit and attracting customers. Incorrect pricing can reduce sales or decrease profitability.

The purpose of this project is to develop a machine learning model that predicts the new_price of perfumes using historical product data. The dataset contains several perfume-related features such as brand, old price, bottle size (ml), concentration type, department, scents, ratings, and seller ratings.

Machine learning helps businesses automate pricing decisions by learning hidden relationships between product features and price. This project demonstrates the practical use of supervised learning techniques in the e-commerce domain.

2.Objectives of the Project

The main objectives of this project are:

- To analyze the perfume dataset and understand important variables.
- To clean and preprocess raw data for machine learning.
- To perform exploratory data analysis (EDA) using charts and graphs.
- To create useful new features from existing data.
- To train multiple regression models.
- To compare model performance and select the best model.
- To predict perfume selling prices accurately.

3.Dataset Description

The dataset used in this project contains perfume products collected from an e-commerce source. Each row represents one perfume product.

Important Columns:

- Brand – Name of perfume brand
- Old_price – Original listed price
- New_price – Discounted/current selling price (Target Variable)
- Ml – Bottle size in milliliters
- Concentration – EDT, EDP, etc.
- Department – Men, Women, Unisex
- Scents – Fragrance category
- Item_rating – Product rating
- Seller_rating – Seller rating
- Num_seller_ratings – Number of seller reviews
- The target variable selected for prediction is new_price.

4.Data Preprocessing

Raw data usually contains noise, irrelevant columns, and inconsistent values. Therefore preprocessing was necessary before model building.

Steps Performed:

a) Removing Unnecessary Columns

Columns such as product name, seller name, and note-related columns were removed because they were less useful for prediction.

b) Handling Missing Values

Missing values were checked and handled appropriately.

c) Outlier Removal

Outliers in numerical columns such as:

- Old_price
- New_price
- Ml
- Were removed using the IQR (Interquartile Range) Method.

This improved model stability and reduced the effect of extreme values.

d) Encoding Categorical Variables

Machine learning models require numerical input. Therefore categorical columns like:

- Brand
- Concentration
- Department
- Scents

Were converted using Label Encoding.

e) Feature Scaling

Numerical columns were standardized using StandardScaler to normalize feature values.

5.Exploratory Data Analysis (EDA)

EDA was performed to understand patterns, distributions, and relationships in the data.

Visualizations Used:

- Histograms
- Boxplots
- Countplots
- Scatterplots
- Correlation Heatmap
- Key Findings:
- Old_price has strong positive correlation with new_price.
- Products with larger ml sizes usually cost more.
- Premium brands tend to have higher selling prices.
- Discounts significantly affect final prices.
- Ratings have smaller influence compared to price-related features.
- EDA helped understand which features are more useful for prediction.

6.Feature Engineering

Feature engineering improves model performance by creating meaningful new variables.

Features Created:

a) Discount_percent

Calculated as:

((old_price – new_price) / old_price) × 100

This feature shows how much discount is offered.

b) Price_per_ml

Calculated as:

New_price / ml

This measures product value relative to bottle size.

These engineered features helped the model capture pricing behavior more effectively.

7.Model Building

Since the target variable new_price is continuous numerical data, regression models were used.

Models Applied:

1. *Linear Regression*

Used as a baseline model to understand linear relationships.

2. *Decision Tree Regressor*

Useful for capturing non-linear price patterns.

3. *Random Forest Regressor*

An ensemble model using multiple trees to improve prediction accuracy.

8.Model Evaluation

Models were evaluated using:

- R^2 Score
- MAE (Mean Absolute Error)
- RMSE (Root Mean Squared Error)
- Final Comparison:
- Linear Regression performed reasonably well.
- Decision Tree captured complex relationships but slightly overfit.
- Random Forest achieved best overall performance.

Best Model:

- Random Forest Regressor
- Because it provided:
- Highest R^2 score
- Lowest error values
- Better generalization

9.Conclusion

This project successfully applied machine learning techniques to predict perfume prices using product features.

The dataset was cleaned, preprocessed, analyzed, and transformed using feature engineering. Multiple regression models were trained and compared.

Among all tested models, Random Forest Regressor performed best and gave the most accurate predictions.

This project shows how machine learning can help e-commerce businesses optimize pricing strategies and improve decision-making.

10.Future Improvements

This project can be improved further by:

Applying GridSearchCV for hyperparameter tuning

Using advanced models like XGBoost

Adding customer reviews and popularity features

Building a Streamlit Web App for real-time price prediction

Using larger real-world datasets

11.Final Statement

Machine learning provides powerful solutions for price prediction problems. This perfume price prediction project demonstrates practical implementation of data preprocessing, EDA, feature engineering, model building, and evaluation in a real-world e-commerce scenario.